UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

John von Neumann was a brilliant mathematician, physicist, and computer scientist. He made groundbreaking contributions to mathematics, game theory, quantum mechanics, and computer science. Von Neumann's work laid the foundation for modern computing, and his ideas continue to shape the fields of mathematics and computer science today.

a) Name the TWO (2) prominent machines that were inspired by John von Neumann during the first computer generation.

b) What is the name of the machine developed by John von Neumann himself?

c) Explain the function of the machine developed by John von Neumann.

QUESTION 2 (5 marks)

a) Discuss TWO (2) major technologies in the fourth computer generation and their impact on the modern computer.

b) Give ONE (1) example of using artificial intelligence (A1) in everyday life.

QUESTION 3 (5 marks)

a) Define byte.

b) Why base 10 is not suitable for computer number representation?

c) In IEEE floating point representation, why exponent is having maximum value of 127 instead of 128?

QUESTION 4 (5 marks)

Question 1 is based on the following Figure 1.
The development of component X led to a drastic increase in speed and efficiency of computers. Component X is a single, complete electronic semiconductor circuits contained on pieces of silicon, and sometimes called as chips.

a) What is the name of component X?

b) In which computer generation was component X introduced?

c) Who is the inventor of component X?

d) State ONE (1) input device and ONE (1) output device that was introduced in the same generation as component X.

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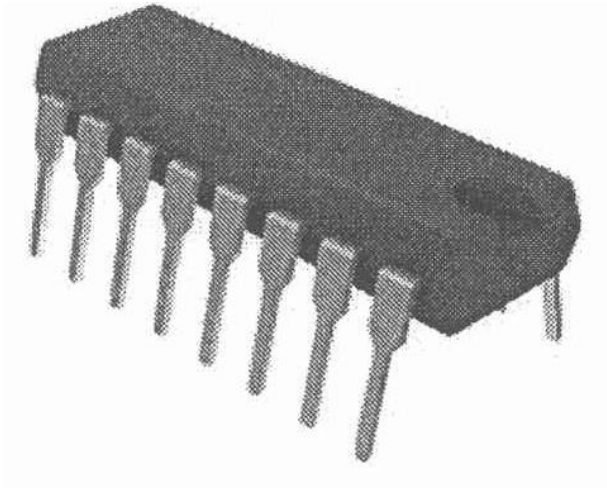


Figure 1: Component X

QUESTION 5 (10 marks)

a) Draw Multiple Instruction, Multiple Data Stream (MIMD) with distributed memory architecture. Use CU as the Control Unit, PU as the Processing Unit, and LM as Local Memory.

b) Draw Multiple Instruction, Multiple Data Stream (MIMD) with shared memory architecture. Use CU as the Control Unit and PU as the Processing Unit.

c) Describe what will happen if the virus attacks the memory in:

   i. Multiple Instruction, Multiple Data Stream (MIMD) with distributed memory.

   ii. Multiple Instruction, Multiple Data Stream (MIMD) with shared memory.

d) Give ONE (1) example of Multiple Instruction, Multiple Data Stream (MIMD) with distributed
memory.

QUESTION 6 (10 marks)

Describe FOUR (4) classifications of multiple processor architecture with an example each.

QUESTION 7 (10 marks)

Given that the size of the data bus is 32 bits, the size of the address bus is 8 bits, and the size of the control bus is 4 bits.

a) Determine the size of the word and provide relevant justification.

b) Determine the size of the register and provide relevant justification.

c) Discuss what will happen if system interconnection is blocked from the perspective of the machine cycle.

d) Elaborate the role of CPU interconnection.

QUESTION 8 (10 marks)

a) Explain why binary number representation used as machine language.

b) Calculate ACEh- BFh in:

i. Hexadecimal

ii. Octal

iii. Binary

QUESTION 9 (10 marks)

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of keyboard characters.

b) Describe THREE (3) categories of input and output module.
Give an appropriate
example for each category.

QUESTION 10 (10 marks)

a) What is a cache?

b) Elaborate on what will happen if there is no cache in the computer system,

c) Comparing L1 and L2 cache, discuss which cache would have faster speed with relevant justification.

d) Discuss TWO (2) policies to write data to the cache to ensure consistency between the cache and the main memory (RAM).

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PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 2 and Table 1.
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 2 using one-, two- and three-address instructions. The instructions available for use are provided in Table 1. Note that Register A, B, C, D, E and F cannot be modified and Register X, Y and Z can be used as temporary register. Final answer must be stored in Register X.

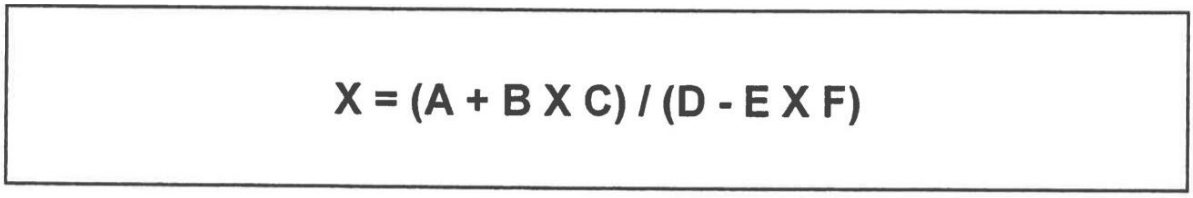


Figure 2

Table 1 : Machine Instruction

i Nia dress		**address**
LOAD M	MOVE (X e Y)	MOVE (X e Y)
STORE M	ADD (X k.. X + Y)	ADD (X e Y + Z)
ADD M	SUB (X e X - Y)	SUB (X e Y-Z)
SUB M	MUL (X e X x Y)	MUL (X e Y x Z)
MUL M	DIV (X e x / Y)	DIV (X e Y / Z)
DIV M		